

WEEK3-

1.SPRING-REST-HANDSON-

Handson1: Create Spring Web Project

```
startedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 21.0.8 with PI
startedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile:
startedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.a
startedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the '
startedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
startedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
startedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
startedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 3
startedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path '/'
startedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 0.774 seconds (process
startedMain] c.c.spring_learn.SpringLearnApplication : end
```

Project

☐ Gradle - Groovy

☐ Gradle - Kotlin ☒ Maven

Spring Boot

☐ 4.1.1 (SNAPSHOT) ☒ 4.1.0 ☐ 4.0.8 (SNAPSHOT) ☐ 4.0.7

☐ 3.5.16

Project Metadata

Group

Artifact

Package name

Packaging ☒ Jar ☐ War

Configuration ☒ Properties ☐ YAML

Java ☐ 26 ☐ 25 ☒ 21 ☐ 17

Language

☒ Java ☐ Kotlin

☐ Groovy

Dependencies

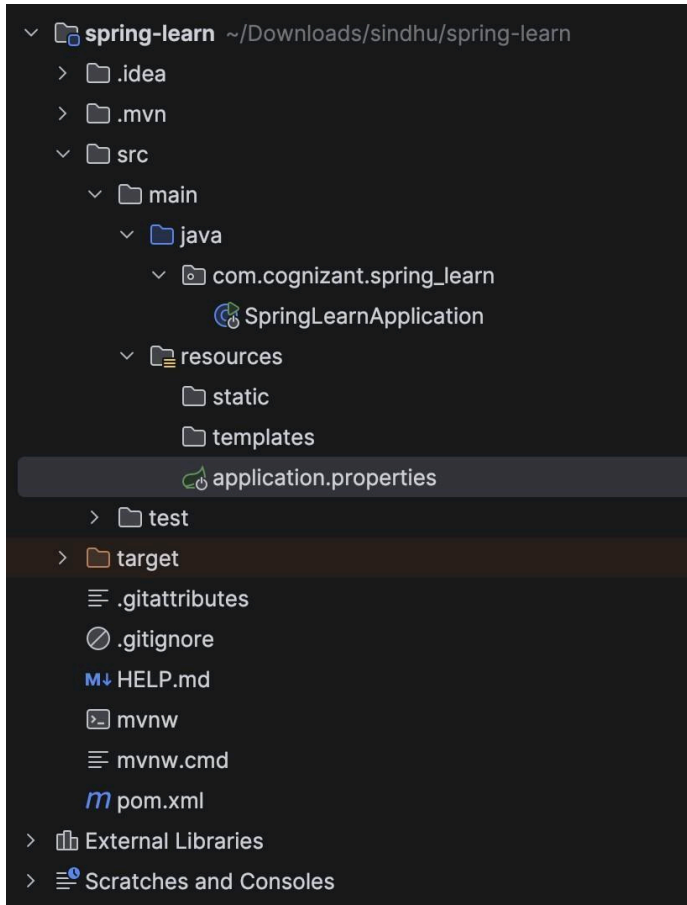
ADD DEPENDENCIES...  + B

Spring Web WEB

Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

Spring Boot DevTools DEVELOPER TOOLS

Provides fast application restarts, LiveReload, and configurations for enhanced development experience.



```
package com.cognizant.spring_learn;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class SpringLearnApplication {
    private static final Logger LOGGER = LoggerFactory.getLogger(SpringLearnApplication.class); 2 usag
    public static void main(String[] args) {
        LOGGER.info("start");

        SpringApplication.run(SpringLearnApplication.class, args);

        LOGGER.info("end");
    }
}
```

HANDSON-2:

```
2026-07-08T13:00:45.395+05:30 INFO 31377 --- [spring-learn] [ re
2026-07-08T13:00:45.398+05:30 INFO 31377 --- [spring-learn] [ re
Mon Dec 31 00:00:00 IST 2018
```

```
SpringLearnApplicationTests.java SpringLearnApplication.java date-format.xml
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
         http://www.springframework.org/schema/beans
         https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="dateFormat"
          class="java.text.SimpleDateFormat">

        <constructor-arg value="dd/MM/yyyy"/>

    </bean>

</beans>
```

```
@SpringBootApplication
public class SpringLearnApplication {

    public static void main(String[] args) throws Exception {

        SpringApplication.run(SpringLearnApplication.class, args);

        displayDate();
    }

    public static void displayDate() throws Exception { 1 usage

        ApplicationContext context =
            new ClassPathXmlApplicationContext( configLocation: "date-format.xml");

        SimpleDateFormat format =
            context.getBean( name: "dateFormat", SimpleDateFormat.class);

        Date date = format.parse( source: "31/12/2018");

        System.out.println(date);
    }
}
```

HANDSON-3:

Spring Core - Incorporate Logging:

```
tion|          logStarting|Starting SpringLearnApplication using Java 21.0.8 with PID 31543 (/Users/sid
tion|    logStartupProfileInfo|No active profile set, falling back to 1 default profile: "default"
ssor|          log|Devtools property defaults active! Set 'spring.devtools.add-properties' to '
ssor|          log|For additional web related logging consider setting the 'logging.level.web'
er. |    initialize|Tomcat initialized with port 8083 (http)
ce. |          log|Starting service [Tomcat]
gine|          log|Starting Servlet engine: [Apache Tomcat/11.0.22]
izer|prepareWebApplicationContext|Root WebApplicationContext: initialization completed in 351 ms
er. |    start|Tomcat started on port 8083 (http) with context path '/'
tion|          logStarted|Started SpringLearnApplication in 0.718 seconds (process running for 1.259)
tion|    displayDate|START
tion|    displayDate|END
```

```

1 import java.util.Date;
2
3 @SpringBootApplication
4 public class SpringLearnApplication {
5
6     private static final Logger LOGGER = 3 usages
7         LoggerFactory.getLogger(SpringLearnApplication.class);
8
9     public static void main(String[] args) throws Exception {
10
11         SpringApplication.run(SpringLearnApplication.class, args);
12
13         displayDate();
14     }
15
16     public static void displayDate() throws Exception { 1 usage
17
18         LOGGER.info("START");
19
20         ApplicationContext context =
21             new ClassPathXmlApplicationContext( configLocation: "date-format.xml");
22
23         SimpleDateFormat format =
24             context.getBean( name: "dateFormat", SimpleDateFormat.class);
25
26         Date date = format.parse( source: "31/12/2018");
27
28         LOGGER.debug("Date : {}", date);
29
30         LOGGER.info("END");
31     }
32 }

```

```

spring.application.name=spring-learn ✓
server.port=8083

logging.level.org.springframework=info
logging.level.com.cognizant.springlearn=debug

logging.pattern.console=%d{yyMMdd} | %d{HH:mm:ss.SSS} | %-20.20thread | %5p | %-25.25logger{25} | %25M | %m%n

```

HANDSON-4:

```
| INFO|ertyDefaultsPostProcessor|           log|Devtools property defaults active! Set 'spring.devtools.add-
| INFO|ertyDefaultsPostProcessor|           log|For additional web related logging consider setting the 'log
| INFO|o.s.b.t.TomcatWebServer|           initialize|Tomcat initialized with port 8083 (http)
| INFO|o.a.c.c.StandardService|           log|Starting service [Tomcat]
| INFO|o.a.c.core.StandardEngine|           log|Starting Servlet engine: [Apache Tomcat/11.0.22]
| INFO|licationContextInitializer|prepareWebApplicationContext|Root WebApplicationContext: initialization completed in 4
| INFO|o.s.b.t.TomcatWebServer|           start|Tomcat started on port 8083 (http) with context path '/'
| INFO|.s.SpringLearnApplication|           logStarted|Started SpringLearnApplication in 0.878 seconds (process run
| INFO|.s.SpringLearnApplication|           displayCountry|START
| DEBUG|c.c.s.model.Country|           <init>|Inside Country Constructor.
| DEBUG|c.c.s.model.Country|           setCode|Inside setCode()
| DEBUG|c.c.s.model.Country|           setName|Inside setName()
| DEBUG|.s.SpringLearnApplication|           displayCountry|Country : Country{code='IN', name='India'}
| INFO|.s.SpringLearnApplication|           displayCountry|END
```

```
package com.cognizant.springlearn;

import com.cognizant.springlearn.model.Country;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

import java.text.SimpleDateFormat;
import java.util.Date;

@SpringBootApplication
public class SpringLearnApplication {
    private static final Logger LOGGER = 6 usages
        LoggerFactory.getLogger(SpringLearnApplication.class);

    public static void main(String[] args) throws Exception {
        SpringApplication.run(SpringLearnApplication.class, args);

        displayCountry();
    }
}
```

```

public class Country { 5 usages

    private static final Logger LOGGER = 5 usages
        LoggerFactory.getLogger(Country.class);

    private String code; 3 usages
    private String name; 3 usages

    public Country() { no usages
        LOGGER.debug("Inside Country Constructor.");
    }

    public String getCode() { no usages
        LOGGER.debug("Inside getCode()");
        return code;
    }

    public void setCode(String code) {
        LOGGER.debug("Inside setCode()");
        this.code = code;
    }

    public String getName() { no usages
        LOGGER.debug("Inside getName()");
        return name;
    }

    public void setName(String name) {
        LOGGER.debug("Inside setName()");
        this.name = name;
    }
}

```

```

<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
        https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="country"
        class="com.cognizant.springlearn.model.Country">

        <property name="code" value="IN"/>
        <property name="name" value="India"/>

    </bean>

</beans>

```

HANDSON:5-

```
ication|          logStarted|Started SpringLearnApplication in 0.813 seconds (pro
ication|          displayCountry|START
/        |          <init>|Inside Country Constructor.
/        |          setCode|Inside setCode()
/        |          setName|Inside setName()
ication|          displayCountry|Country 1 : Country{code='IN', name='India'}
ication|          displayCountry|Country 2 : Country{code='IN', name='India'}
ication|          displayCountry|END
```

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="country"
          class="com.cognizant.springlearn.model.Country">

        <property name="code" value="IN"/>
        <property name="name" value="India"/>

    </bean>

</beans>
```

```

displayCountry|START
    <init>|Inside Country Constructor.
    setCode|Inside setCode()
    setName|Inside setName()
    <init>|Inside Country Constructor.
    setCode|Inside setCode()
    setName|Inside setName()
displayCountry|Country 1 : Country{code='IN', name='India'}
displayCountry|Country 2 : Country{code='IN', name='India'}
displayCountry|END

```

```

}
public static void displayCountry() { 1 usage

    LOGGER.info("START");

    ApplicationContext context =
        new ClassPathXmlApplicationContext( configLocation: "country.xml");

    Country country =
        context.getBean( name: "country", Country.class);

    Country anotherCountry =
        context.getBean( name: "country", Country.class);

    LOGGER.debug("Country 1 : {}", country);
    LOGGER.debug("Country 2 : {}", anotherCountry);

    LOGGER.info("END");
}
}

```

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
         http://www.springframework.org/schema/beans
         https://www.springframework.org/schema/beans/spring-beans.xsd">

  <bean id="country"
        class="com.cognizant.springlearn.model.Country"
        scope="prototype">

    <property name="code" value="IN"/>
    <property name="name" value="India"/>

  </bean>

</beans>
```